

King Fahd University of Petroleum and Minerals
Department of Mathematics

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Math 208 Major Exam I Term 252

Original exam date: 16 February 2026. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. The solution of the initial-value problem $(2 - \cos x)y' + y \sin x = 2 + \cos x$, $y(0) = 0$ is

(a) $y = \frac{2x + \cos x}{2 + \cos x}$

(b) $y = \frac{x + \cos x}{2 + \cos x}$

(c) $y = \frac{x + \sin x}{2 - \cos x}$

(d) $y = \frac{2x + \sin x}{2 - \cos x}$

(e) $y = \frac{2x + \sin x}{2x - \cos x}$

2. The sum of all real values of m that make the equation $(mx^2y + 2xy - y^3)dx + (x^2 - 3xy^2 + \frac{1}{3}m^3x^3)dy = 0$ an exact differential equation is equal to

(a) 1

(b) -1

(c) 0

(d) -2

(e) 2

3. A particle is moving in a straight line with acceleration $a(t) = \ln(5t)$ and initial velocity $v(\frac{1}{5}) = \frac{4}{5}$. The velocity at any time $t > \frac{1}{5}$ is given by

(a) $v(t) = 2t \ln(5t) - t + 1$

(b) $v(t) = t \ln(5t) + 2t + \frac{2}{5}$

(c) $v(t) = t \ln(5t) - t + 1$

(d) $v(t) = 3t \ln(5t) - t + 1$

(e) $v(t) = t \ln(5t) - 2t + \frac{6}{5}$

4. The general solution of the differential equation $x \frac{dy}{dx} = y + 4xe^{-2y/x}$ for $x \neq 0$ is given by

(a) $e^{2x/y} + 4 \ln |x| = c$

(b) $e^{2y/x} - 8 \ln |x| = c$

(c) $e^{2x/y} + 8 \ln |x| = c$

(d) $e^{2y/x} - \ln |x| = c$

(e) $e^{2y/x} - 4 \ln |x| = c$

5. The general solution of the differential equation $\frac{dy}{dx} = \frac{x + y + 2}{2x + 2y - 1}$ is given by

(a) $2x + y + \frac{5}{3} \ln |3x + 3y - 1| = c$

(b) $2x + 2y - \frac{5}{3} \ln |2x + 2y - 1| = c$

(c) $2y + x - \frac{5}{3} \ln |3x + 3y + 1| = c$

(d) $2y - x - \frac{5}{3} \ln |3x + 3y + 1| = c$

(e) $x + y + \frac{5}{3} \ln |3x + 3y - 1| = c$

6. The general solution of the exact differential equation $(2x + y^2 e^{xy}) dx + (e^{xy} + xy e^{xy}) dy = 0$ is

(a) $ye^{xy} + x^3 = c$

(b) $ye^{xy} + y^2 = c$

(c) $ye^{xy} + x^2 = c$

(d) $xe^{xy} + y^2 = c$

(e) $xe^{xy} + x^2 = c$

7. The sum of all real values of k for which the vectors $\mathbf{v}_1 = (2, 3, 0)$, $\mathbf{v}_2 = (4, k, 4)$, $\mathbf{v}_3 = (-1, 1, k)$ do not form a basis for $\mathbb{R}^3$ is

(a) 6

(b) 4

(c) 5

(d) -12

(e) 12

8. By using a suitable substitution, we can transform the differential equation $y' + \frac{2}{x}y = 4x^3y^4$ into the linear differential equation

(a) $\frac{dw}{dx} - \frac{2}{x}w = 4x^3$

(b) $\frac{dw}{dx} - \frac{6}{x}w = 6x^3$

(c) $\frac{dw}{dx} - \frac{3}{x}w = 6x^3$

(d) $\frac{dw}{dx} - \frac{6}{x}w = -12x^3$

(e) $\frac{dw}{dx} - \frac{3}{x}w = -12x^3$

9. Which one of the following statements is true about the subset V of $\mathbb{R}^3$ defined by $V = \{(x_1, x_2, x_3) : x_3 = x_1 + x_2\}$?

(a) V is a subspace of $\mathbb{R}^3$

(b) V is not closed under addition but closed under multiplication by scalar

(c) V is not closed under addition

(d) V is closed under addition but not closed under multiplication by scalars

(e) V is not closed under addition but not closed under multiplication by scalars

10. All values of m such that $y = e^{mx}$ is a solution of the differential equation $y'' - 6y' - 7y = 0$ are

(a) $m = 7$ and $m = 2$

(b) $m = -7$ and $m = 2$

(c) $m = 7$ and $m = 1$

(d) $m = 7$ and $m = -2$

(e) $m = 7$ and $m = -1$

11. A certain radioactive substance decomposes at a rate directly proportional to the quantity present. If 50% of a certain quantity has decomposed in 20 years, how many years will it take for 80% of the original quantity to decompose?

(a) $\frac{20 \ln 5}{\ln 2}$

(b) $\frac{15 \ln 5}{\ln 2}$

(c) $\frac{30 \ln 5}{\ln 2}$

(d) $\frac{24 \ln 5}{\ln 2}$

(e) $\frac{40 \ln 5}{\ln 2}$

12. The order of the differential equation $y''' + 2x(y'')^4 + x^2(y')^6 = 0$ is

- (a) 1
- (b) 6
- (c) 10
- (d) 4
- (e) 3

13. By using a suitable substitution, we can transform the differential equation $yy'' + (y')^2 = 0$ into the separable differential equation

- (a) $\frac{dp}{2p} + \frac{dy}{y} = 0$
- (b) $\frac{dp}{2p} - \frac{dy}{y} = 0$
- (c) $\frac{dp}{p} - \frac{dy}{3y} = 0$
- (d) $\frac{dp}{p} + \frac{dy}{2y} = 0$
- (e) $\frac{dp}{p} + \frac{dy}{y} = 0$

14. The general solution of the differential equation $\frac{dy}{dx} = x\sqrt{\frac{1-y^2}{1-x^2}}$ for $|x| < 1$ and $|y| < 1$ is given by

- (a) $y = 3 \sin \left(c - \sqrt{1-x^2} \right)$
- (b) $y = \frac{1}{3} \sin \left(c - \sqrt{1-x^2} \right)$
- (c) $y = \sin \left(c - \sqrt{1-x^2} \right)$
- (d) $y = \frac{1}{2} \sin \left(c - \sqrt{1-x^2} \right)$
- (e) $y = 2 \sin \left(c - \sqrt{1-x^2} \right)$

15. The solution of the system

$$x_1 + 3x_2 + 3x_3 + 3x_4 = 0$$

$$2x_1 + 7x_2 + 5x_3 - x_4 = 0$$

$$2x_1 + 7x_2 + 4x_3 - 4x_4 = 0$$

is the set of all scalar multiples of a vector $\mathbf{u} = (a, b, c, d)$. Then $\frac{a}{c} =$

(a) -2

(b) 4

(c) -4

(d) 0

(e) 2

Answer Key

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Q	Ans	Brief explanation
1	(d)	$\rho = 2 - \cos x$: $[(2 - \cos x)y]' = 2 + \cos x$, $C = 0$.
2	(c)	$M_y = N_x$ gives $m = m^3$: $m = 0, \pm 1$.
3	(c)	$v = t \ln(5t) - t + C$; $v(\frac{1}{5}) = \frac{4}{5}$ gives $C = 1$.
4	(b)	$v = y/x$: $e^{2v} dv = \frac{4}{x} dx$, $\frac{1}{2} e^{2v} = 4 \ln x + C$.
5	(d)	$v = x + y$: $\frac{2v-1}{3v+1} dv = dx$.
6	(c)	$F = x^2 + ye^{xy} + g(y)$ with $g' = 0$.
7	(a)	$\det = 2k^2 - 12k - 20 = 0$; roots sum to 6.
8	(d)	$w = y^{-3}$: $-\frac{1}{3}w' + \frac{2}{x}w = 4x^3$.
9	(a)	$x_3 = x_1 + x_2$ is homogeneous linear, so both closures hold.
10	(e)	$m^2 - 6m - 7 = (m - 7)(m + 1) = 0$.
11	(a)	$k = \frac{\ln 2}{20}$; 20% left: $kt = \ln 5$.
12	(e)	Highest derivative is y''' .
13	(e)	x missing: $p = y'$, $y'' = p \frac{dp}{dy}$, divide by yp^2 .
14	(c)	Separable: $\sin^{-1} y = -\sqrt{1 - x^2} + c$.
15	(e)	$\mathbf{u} = (-6, 4, -3, 1)$, so $a/c = 2$.